

Linguaggio Fun e le relazioni ‘ $\rightarrow$ ’ e ‘ $\rightsquigarrow$ ’

I termini del linguaggio sono generati dalla seguente grammatica

$$t ::= \mathbf{c} \mid x \mid \mathbf{op}_i(t_1, \dots, t_{a_i}) \mid \mathbf{if}(t_1, t_2, t_3) \mid \mathbf{fun}(x, t) \mid \mathbf{apply}(t_1, t_2) \mid \mathbf{let} \ x = t_1 \ \mathbf{in} \ t_2$$

dove le $\mathbf{op}_i$ sono, ad esempio, *sum*, *and* o *uguale* e $\mathbf{c}$ sono le costanti, ad esempio 1, 2 o *true*.

Le regole per la relazione *big-step* sono nella figura qui sotto

$$\begin{array}{c}
 \frac{}{\langle x, \delta \rangle \rightarrow \delta(x)} \qquad \frac{}{\langle \mathbf{c}, \delta \rangle \rightarrow \mathbf{c}} \qquad \frac{}{\langle \mathbf{fun}(x, t), \delta \rangle \rightarrow \mathbf{fun}(x, t)} \\
 \\
 \frac{\langle t_1, \delta \rangle \rightarrow \mathbf{v}_1 \quad \dots \quad \langle t_{a_i}, \delta \rangle \rightarrow \mathbf{v}_{a_i}}{\langle \mathbf{op}_i(t_1, \dots, t_{a_i}), \delta \rangle \rightarrow \widetilde{op}_i(\mathbf{v}_1, \dots, \mathbf{v}_{a_i})} \qquad \frac{\langle t_0, \delta \rangle \rightarrow \mathbf{true} \quad \langle t_1, \delta \rangle \rightarrow \mathbf{v}}{\langle \mathbf{if} \ t_0 \ \mathbf{then} \ t_1 \ \mathbf{else} \ t_2, \delta \rangle \rightarrow \mathbf{v}} \\
 \\
 \frac{\langle t_0, \delta \rangle \rightarrow \mathbf{false} \quad \langle t_2, \delta \rangle \rightarrow \mathbf{v}}{\langle \mathbf{if} \ t_0 \ \mathbf{then} \ t_1 \ \mathbf{else} \ t_2, \delta \rangle \rightarrow \mathbf{v}} \qquad \frac{\langle t_1, \delta \rangle \rightarrow \mathbf{fun}(x, t) \quad \langle t_2, \delta \rangle \rightarrow \mathbf{v}' \quad \langle t, \delta[x/\mathbf{v}_2] \rangle \rightarrow \mathbf{v}}{\langle \mathbf{apply}(t_1, t_2), \delta \rangle \rightarrow \mathbf{v}} \\
 \\
 \frac{\langle t_1, \delta \rangle \rightarrow \mathbf{v}_1 \quad \langle t_2, \delta[x/\mathbf{v}_1] \rangle \rightarrow \mathbf{v}}{\langle \mathbf{let} \ x = t_1 \ \mathbf{in} \ t_2, \delta \rangle \rightarrow \mathbf{v}}
 \end{array}$$

Figura 1: La relazione big-step ‘ $\rightarrow$ ’

Un esempio di derivazione è il seguente. Considerate il termine $t = \mathbf{let} \ f = \mathbf{fun}(x, x) \ \mathbf{in} \ \mathbf{apply}(f, 1)$. La valutazione usando la big-step semantics è data dal seguente albero di derivazione, dove sono indicati i risultati intermedi e gli ambienti

$$\frac{\frac{}{\langle \mathbf{fun}(x, x), \perp \rangle \rightarrow \mathbf{fun}(x, x)} \quad \frac{\frac{\langle f, \delta \rangle \rightarrow \delta(f) = \mathbf{fun}(x, x)}{\langle \mathbf{fun}(x, x), \perp \rangle \rightarrow \mathbf{fun}(x, x)} \quad \frac{\langle 1, \delta \rangle \rightarrow 1}{\langle \mathbf{apply}(f, 1), \delta \rangle \rightarrow 1} \quad \frac{\langle x, \delta' \rangle \rightarrow \delta'(x) = 1}{\langle \mathbf{apply}(f, 1), \delta \rangle \rightarrow 1}}{\langle \mathbf{let} \ f = \mathbf{fun}(x, x) \ \mathbf{in} \ \mathbf{apply}(f, 1), \perp \rangle \rightarrow 1}$$

dove $\delta = \perp[f/\mathbf{apply}(f, 1)]$ e $\delta' = \delta[x/1]$

Nella figura 2 sono elencate le regole per la *small-step semantics*

$$\begin{array}{c}
\frac{}{\langle\langle x, \Delta \rangle\rangle \rightsquigarrow \langle\text{top}(\Delta)(x), \Delta\rangle} \quad (1) \qquad \frac{}{\langle\langle \text{op}_i(v_1, \dots, v_{a_i}), \Delta \rangle\rangle \rightsquigarrow \langle\langle v, \Delta \rangle\rangle} \quad (2) \\
\\
\frac{\langle\langle t_j, \Delta \rangle\rangle \rightsquigarrow \langle\langle t'_j, \Delta' \rangle\rangle}{\langle\langle \text{op}_i(v_1, \dots, v_{j-1}, t_j, \dots, t_{a_i}), \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{op}_i(v_1, \dots, v_{j-1}, t'_j, \dots, t_{a_i}), \Delta' \rangle\rangle} \quad (3) \\
\\
\frac{}{\langle\langle \text{if true then } t_1 \text{ else } t_2, \Delta \rangle\rangle \rightsquigarrow \langle\langle t_1, \Delta \rangle\rangle} \quad (4) \qquad \frac{}{\langle\langle \text{if false then } t_1 \text{ else } t_2, \Delta \rangle\rangle \rightsquigarrow \langle\langle t_2, \Delta \rangle\rangle} \quad (5) \\
\\
\frac{\langle\langle t_0, \Delta \rangle\rangle \rightsquigarrow \langle\langle t'_0, \Delta' \rangle\rangle}{\langle\langle \text{if } t_0 \text{ then } t_1 \text{ else } t_2, \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{if } t'_0 \text{ then } t_1 \text{ else } t_2, \Delta' \rangle\rangle} \quad (6) \\
\\
\frac{}{\langle\langle \text{apply}(\text{fun}(x, t), v), \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{funblock}(t), \text{push}(\text{top}(\Delta)[x/v], \Delta) \rangle\rangle} \quad (7) \\
\\
\frac{\langle\langle t_1, \Delta \rangle\rangle \rightsquigarrow \langle\langle t'_1, \Delta' \rangle\rangle}{\langle\langle \text{apply}(t_1, t_2), \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{apply}(t'_1, t_2), \Delta' \rangle\rangle} \quad (8) \\
\\
\frac{\langle\langle t_2, \Delta \rangle\rangle \rightsquigarrow \langle\langle t'_2, \Delta' \rangle\rangle}{\langle\langle \text{apply}(\text{fun}(x, t), t_2), \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{apply}(\text{fun}(x, t), t'_2), \Delta' \rangle\rangle} \quad (9) \\
\\
\frac{\langle\langle t, \Delta \rangle\rangle \rightsquigarrow \langle\langle t', \Delta' \rangle\rangle}{\langle\langle \text{funblock}(t), \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{funblock}(t'), \Delta' \rangle\rangle} \quad (10) \qquad \frac{}{\langle\langle \text{funblock}(v), \Delta \rangle\rangle \rightsquigarrow \langle\langle v, \text{pop}(\Delta) \rangle\rangle} \quad (11) \\
\\
\frac{\langle\langle t_1, \Delta \rangle\rangle \rightsquigarrow \langle\langle t'_1, \Delta' \rangle\rangle}{\langle\langle \text{let } x = t_1 \text{ in } t_2, \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{let } x = t'_1 \text{ in } t_2, \Delta' \rangle\rangle} \quad (12) \\
\\
\frac{}{\langle\langle \text{let } x = v \text{ in } t, \Delta \rangle\rangle \rightsquigarrow \langle\langle \text{funblock}(t), \text{push}(\text{top}(\Delta)[x/v], \Delta) \rangle\rangle} \quad (13)
\end{array}$$

Figura 2: La relazione small-step ‘ $\rightsquigarrow$ ’

Un esempio di derivazione (del termine $t = \text{let } f = \text{fun}(x, x) \text{ in apply}(f, 1)$) è la seguente, e la pila vuota è indicata con *emptystack*

	regola applicata
$\langle\langle \text{let } f = \text{fun}(x, x) \text{ in apply}(f, 1), \Delta \rangle\rangle$	$\rightsquigarrow$ (13)
$\langle\langle \text{funblock}(\text{apply}(f, 1)), \text{push}(\text{top}(\Delta)[f/\text{fun}(x, x)], \Delta) \rangle\rangle$	$\rightsquigarrow$ (8)
<i>questa usa la regola (1) : $\langle\langle f, \Delta' \rangle\rangle \rightsquigarrow \langle\langle \text{top}(\Delta')(f), \Delta' \rangle\rangle$ dove $\text{fun}(x, x) = \text{top}(\Delta')(f)$</i>	
$\langle\langle \text{funblock}(\text{apply}(\text{fun}(x, x), 1)), \Delta' \rangle\rangle$	$\rightsquigarrow$ (7)
$\langle\langle \text{funblock}(\text{funblock}(x)), \text{push}(\text{top}(\Delta')[x/1], \Delta') \rangle\rangle$	$\rightsquigarrow$ (10)
<i>questa usa la regola (1) : $\langle\langle x, \Delta'' \rangle\rangle \rightsquigarrow \langle\langle \text{top}(\Delta'')(x), \Delta'' \rangle\rangle$ dove $1 = \text{top}(\Delta'')(x)$</i>	
$\langle\langle \text{funblock}(\text{funblock}(1)), \Delta'' \rangle\rangle$	$\rightsquigarrow$ (11)
$\langle\langle \text{funblock}(1), \Delta' \rangle\rangle$	$\rightsquigarrow$ (11)
$\langle\langle 1, \Delta \rangle\rangle$	

dove $\Delta = \text{push}(\perp, \text{emptystack})$, $\Delta' = \text{push}(\text{top}(\Delta)[f/\text{fun}(x, x)], \Delta)$ e $\Delta'' = \text{push}(\text{top}(\Delta')[x/1], \Delta')$

nell'applicazione delle regole è indicata non solo la regola usata ma anche la derivazione che ha determinato la premessa

Un linguaggio funzionale Fun_τ e le regole per l'inferenza di tipo

Il linguaggio considerato è generato dalla seguente grammatica: $t ::= x \mid n \mid true \mid false \mid t_1 + t_2 \mid t_1 - t_2 \mid t_1 \times t_2 \mid t_1 \wedge t_2 \mid t_1 \vee t_2 \mid \neg t_1 \mid t_1 = t_2 \mid t_1 < t_2 \mid if\ t_0\ then\ t_1\ else\ t_2 \mid fun(x, t) \mid apply(t_1, t_2) \mid pair(t_1, t_2) \mid fst(t) \mid snd(t)$ mentre i tipi sono $\tau ::= a \mid int \mid bool \mid \tau_1 \rightarrow \tau_2 \mid \tau_1 * \tau_2$ dove a sono le variabili di tipo. Infine Γ è un insieme di coppie (x, τ) dove x è una variabile del linguaggio e τ un tipo.

$\frac{(x, \tau) \in \Gamma}{\Gamma \vdash x : \tau}$	$\frac{}{\Gamma \vdash n : int}$	$\frac{}{\Gamma \vdash true : bool}$	$\frac{}{\Gamma \vdash false : bool}$
$\frac{\Gamma \vdash t_1 : int \quad \Gamma \vdash t_2 : int}{\Gamma \vdash t_1\ op\ t_2 : int} \quad op \in \{+, -, \times\}$	$\frac{\Gamma \vdash t_1 : \tau \quad \Gamma \vdash t_2 : \tau}{\Gamma \vdash t_1 = t_2 : bool} \quad \tau \neq \tau_1 \rightarrow \tau_2$		
$\frac{\Gamma \vdash t_1 : bool \quad \Gamma \vdash t_2 : bool}{\Gamma \vdash t_1\ op\ t_2 : bool} \quad op \in \{\wedge, \vee\}$	$\frac{\Gamma \vdash t : bool}{\Gamma \vdash \neg t : bool}$	$\frac{\Gamma \vdash t_1 : int \quad \Gamma \vdash t_2 : int}{\Gamma \vdash t_1 < t_2 : bool}$	
$\frac{\Gamma \vdash x : \tau_1 \quad \Gamma \vdash t : \tau_2}{\Gamma \vdash fun(x, t) : \tau_1 \rightarrow \tau_2}$	$\frac{\Gamma \vdash t_0 : bool \quad \Gamma \vdash t_1 : \tau \quad \Gamma \vdash t_2 : \tau}{\Gamma \vdash if\ t_0\ then\ t_1\ else\ t_2 : \tau}$	$\frac{\Gamma \vdash t_1 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash t_2 : \tau_1}{\Gamma \vdash apply(t_1, t_2) : \tau_2}$	
$\frac{\Gamma \vdash t_1 : \tau_1 \quad \Gamma \vdash t_2 : \tau_2}{\Gamma \vdash pair(t_1, t_2) : \tau_1 * \tau_2}$	$\frac{\Gamma \vdash t : \tau_1 * \tau_2}{\Gamma \vdash fst(t) : \tau_1}$	$\frac{\Gamma \vdash t : \tau_1 * \tau_2}{\Gamma \vdash snd(t) : \tau_2}$	

Figura 3: La relazione ‘ $\vdash$ ’ per tipare un termine

Considerate il tipo

$$int * (int \rightarrow bool) \rightarrow bool$$

Un termine nel linguaggio che ha questo tipo potrebbe essere

$$\neg apply(snd(x), fst(x) + 1)$$

Poniamo $\Gamma = \{(x, int * (int \rightarrow bool))\}$, un derivazione che mostra che quel termine ha proprio quel tipo è:

$$\frac{\frac{(x, int * (int \rightarrow bool)) \in \Gamma}{\Gamma \vdash x : int * (int \rightarrow bool)} \quad \frac{\frac{(x, int * (int \rightarrow bool)) \in \Gamma}{\Gamma \vdash x : int * (int \rightarrow bool)} \quad \frac{\Gamma \vdash fst(x) : int \quad \Gamma \vdash 1 : int}{\Gamma \vdash fst(x) + 1 : int}}{\Gamma \vdash snd(x) : int \rightarrow bool} \quad \frac{\Gamma \vdash snd(x) : int \rightarrow bool \quad \Gamma \vdash fst(x) + 1 : int}{\Gamma \vdash apply(snd(x), fst(x) + 1) : bool} \quad \frac{}{\Gamma \vdash \neg apply(snd(x), fst(x) + 1) : bool}$$

La figura nella prossima pagina contiene le regole per inferire il tipo di un termine del linguaggio funzionale Fun_τ .

$\frac{}{\langle\langle x, type \rangle\rangle \Rightarrow (type(x), \emptyset)}$	$\frac{}{\langle\langle n, type \rangle\rangle \Rightarrow (\text{int}, \emptyset)}$
$\frac{\langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle t_1 \text{ op } t_2, type \rangle\rangle \Rightarrow (\text{int}, \{(\tau_1, \text{int}), (\tau_2, \text{int})\} \cup T_1 \cup T_2)} \text{ op} \in \{+, -, \times\}$	$\frac{}{\langle\langle true, type \rangle\rangle \Rightarrow (\text{bool}, \emptyset)}$
$\frac{\langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle t_1 \text{ op } t_2, type \rangle\rangle \Rightarrow (\text{bool}, \{(\tau_1, \text{bool}), (\tau_2, \text{bool})\} \cup T_1 \cup T_2)} \text{ op} \in \{\wedge, \vee\}$	$\frac{}{\langle\langle false, type \rangle\rangle \Rightarrow (\text{bool}, \emptyset)}$
$\frac{\langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle t_1 < t_2, type \rangle\rangle \Rightarrow (\text{bool}, \{(\tau_1, \text{int}), (\tau_2, \text{int})\} \cup T_1 \cup T_2)}$	$\frac{\langle\langle t, type \rangle\rangle \Rightarrow (\tau, T)}{\langle\langle \neg t, type \rangle\rangle \Rightarrow (\text{bool}, \{(\tau, \text{bool})\} \cup T)}$
$\frac{\langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle t_1 = t_2, type \rangle\rangle \Rightarrow (\text{bool}, \{(\tau_1, \tau_2)\} \cup T_1 \cup T_2)}$	$\frac{\langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle pair(t_1, t_2), type \rangle\rangle \Rightarrow (\tau, \{(\tau, \tau_1 * \tau_2)\} \cup T_1 \cup T_2)}$
$\frac{a = \text{newvar}() \quad \langle\langle t, type \rangle\rangle \Rightarrow (\tau, T)}{\langle\langle fst(t), type \rangle\rangle \Rightarrow (\tau_1, \{(\tau, \tau_1 * a)\} \cup T)}$	$\frac{a = \text{newvar}() \quad \langle\langle t, type \rangle\rangle \Rightarrow (\tau, T)}{\langle\langle snd(t), type \rangle\rangle \Rightarrow (\tau_2, \{(\tau, a * \tau_2)\} \cup T)}$
$\frac{\langle\langle t_0, type \rangle\rangle \Rightarrow (\tau_0, T_0) \quad \langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle \text{if } t_0 \text{ then } t_1 \text{ else } t_2, type \rangle\rangle \Rightarrow (\tau, \{(\tau_0, \text{bool}), (\tau, \tau_1), (\tau_1, \tau_2)\} \cup T_0 \cup T_1 \cup T_2)}$	
$\frac{a = \text{newvar}() \quad \langle\langle t, type[x/a] \rangle\rangle \Rightarrow (\tau_2, T)}{\langle\langle fun(x, t), type \rangle\rangle \Rightarrow (a \rightarrow \tau_2, T)}$	
$\frac{a = \text{newvar}() \quad \langle\langle t_1, type \rangle\rangle \Rightarrow (\tau, T_2) \quad \langle\langle t_2, type \rangle\rangle \Rightarrow (\tau_1, T_1)}{\langle\langle apply(t_1, t_2), type \rangle\rangle \Rightarrow (a, \{(\tau, \tau_1 \rightarrow a)\} \cup T_1 \cup T_2)}$	
$\frac{a = \text{newvar}() \quad \langle\langle t_1, type \rangle\rangle \Rightarrow (\tau_1, T_1) \quad \langle\langle t_2, type[x/a] \rangle\rangle \Rightarrow (\tau_2, T_2)}{\langle\langle let x \leftarrow t_1 \text{ in } t_2, type \rangle\rangle \Rightarrow (\tau_2, \{(\tau_1, a)\} \cup T_1 \cup T_2)}$	

Figura 4: Regole per inferire un tipo

Consideriamo il termine $t = \neg apply(snd(x), fst(x) + 1)$

$$\begin{array}{c}
\frac{a = \text{newvar}() \quad \frac{b = \text{newvar}() \quad \langle\langle x, type \rangle\rangle \Rightarrow (\tau_1, \emptyset)}{\langle\langle snd(x), type \rangle\rangle \Rightarrow (\tau, \{(\tau_1, b * \tau)\})} \quad \frac{c = \text{newvar}() \quad \langle\langle x, type \rangle\rangle \Rightarrow (\tau_1, \emptyset)}{\langle\langle fst(x), type \rangle\rangle \Rightarrow (\tau_2, \{(\tau_1, \tau_2 * c)\})} \quad \langle\langle 1, type \rangle\rangle \Rightarrow (\text{int}, \emptyset)}{\langle\langle fst(x) + 1, type \rangle\rangle \Rightarrow (\text{int}, \{(\tau_2, \text{int}), (\text{int}, \text{int}), (\tau_1, \tau_2 * c)\})} \\
\frac{\langle\langle apply(snd(x), fst(x) + 1), type \rangle\rangle \Rightarrow (a, \{(\tau, \text{int} \rightarrow a) \cup \{(\tau_1, b * \tau)\} \cup \{(\tau_2, \text{int}), (\text{int}, \text{int}), (\tau_1, \tau_2 * c)\})\})}{\langle\langle \neg apply(snd(x), fst(x) + 1), type \rangle\rangle \Rightarrow (\text{bool}, \{(a, \text{bool}), (\tau, \text{int} \rightarrow a), (\tau_1, b * \tau), (\tau_2, \text{int}), (\text{int}, \text{int}), (\tau_1, \tau_2 * c)\})}
\end{array}$$

In questa derivazione abbiamo che $type(x) = \tau_1$.

$$\{(a, \text{bool}), (\tau, \text{int} \rightarrow a), (\tau_1, b * \tau), (\tau_2, \text{int}), (\text{int}, \text{int}), (\tau_1, \tau_2 * c)\}$$

Se risolviamo i vincoli, cerchiamo cioè una sostituzione per le incognite che rendono le coppie formate da elementi uguali, otteniamo che $x = \text{int} * (\text{int} \rightarrow \text{bool})$ dato che la sostituzione adatta è $a = \text{bool}$, $b = \text{int}$ e $c = \text{int} \rightarrow \text{bool}$. Quindi il Γ sarà $\{(x, \text{int} * (\text{int} \rightarrow \text{bool}))\}$ che è quello usato prima.